

DriveEase

Data Models Reference

Entities, attributes, keys, sample data and relationships

ASP.NET Core 9.0 MVC | Entity Framework Core | PostgreSQL (Supabase)

Legend:

PK = Primary Key

FK = Foreign Key

? = Nullable / Optional

Models Overview

The domain has **8 entity models** plus a set of enums. Each model maps to a table in the PostgreSQL database via Entity Framework Core.

#	Model	Table Role	Primary Key	Foreign Keys
1	User	People: customers & admins	Id	—
2	Vehicle	Rentable cars & bikes	Id	—
3	Booking	Rental reservations	Id	UserId, VehicleId, PromoCodeId?
4	Payment	Payment per booking	Id	BookingId
5	Review	Ratings & comments	Id	UserId, VehicleId, BookingId
6	PromoCode	Discount coupons	Id	—
7	PricingRule	Dynamic pricing (surge)	Id	—
8	Notification	SMS-style messages	Id	UserId

1. User

Represents an account in the system. A user is either a **Customer** (who rents vehicles) or an **Administrator** (who manages the fleet). Email is uniquely indexed.

Attribute	Data Type	Key	Stores / Description	Example
Id	int	PK	Auto-generated unique identifier for the user.	1
FullName	string(100)		User's full name. Required, 3-100 chars.	Rahul Sharma
Email	string		Login email. Required, unique, validated format.	rahul@gmail.com
MobileNumber	string		Contact / login mobile. Required, phone format.	9823456780
DrivingLicenseNumber	string?		License no. (needed for self-drive). Optional on entity.	MH12-2021-00892
Password	string		Account password (stored as plain text today).	User@123
Role	UserRole		Customer or Administrator.	Customer
IsActive	bool		Whether the account is enabled.	true
CreatedAt	DateTime		When the account was created (UTC).	2026-07-21T10:00Z
Bookings	ICollection<Booking>	nav	Navigation: all bookings by this user.	[...]
Reviews	ICollection<Review>	nav	Navigation: all reviews by this user.	[...]

2. Vehicle

A rentable car or bike in the fleet. LicensePlate is uniquely indexed. Rating fields are cached aggregates recomputed when reviews are added.

Attribute	Data Type	Key	Stores / Description	Example
Id	int	PK	Auto-generated unique identifier.	3
Brand	string(50)		Manufacturer. Required.	Maruti Suzuki
Model	string(50)		Model name. Required.	Swift VXi
Year	int		Manufacturing year.	2023
LicensePlate	string(20)		Registration plate. Required, unique.	MH12-AB-1234
VehicleType	VehicleType		Bike, Hatchback, Sedan, SUV or Luxury.	Hatchback
SeatingCapacity	int		Number of seats (2 bike, 5 sedan).	5
FuelType	FuelType		Petrol, Diesel or Electric.	Petrol
Transmission	TransmissionType		Manual or Automatic.	Manual
HourlyRate	decimal(18,2)		Price per hour (Rs).	150.00
DailyRate	decimal(18,2)		Price per day (Rs).	1800.00
PickupLocation	string(100)		Where vehicle is picked up. Required.	Downtown Hub
ImageUrl	string		Path to vehicle image.	/images/vehicles/swift.png
Status	VehicleStatus		Available, Booked or UnderMaintenance.	Available
AverageRating	decimal(3,2)		Cached average of reviews.	4.60
ReviewCount	int		Cached count of reviews.	12
CreatedAt	DateTime		When added to fleet (UTC).	2026-06-21T10:00Z

Attribute	Data Type	Key	Stores / Description	Example
Bookings	ICollection<Booking>	nav	Navigation: bookings of this vehicle.	[...]
Reviews	ICollection<Review>	nav	Navigation: reviews of this vehicle.	[...]

3. Booking

A rental reservation linking a user to a vehicle for a time window. BookingReference is uniquely indexed. Holds pricing snapshot and status.

Attribute	Data Type	Key	Stores / Description	Example
Id	int	PK	Auto-generated unique identifier.	10
BookingReference	string(30)		Human-readable code. Required, unique.	DE-202609-1002
UserId	int	FK	-> User.Id (who booked).	2
User	User?	nav	Navigation to the booking's user.	{ Rahul }
VehicleId	int	FK	-> Vehicle.Id (what was booked).	3
Vehicle	Vehicle?	nav	Navigation to the booked vehicle.	{ Swift }
RentalType	RentalType		Hourly or Daily.	Daily
PickupDateTime	DateTime		Start of the rental.	2026-09-26T09:00Z
ReturnDateTime	DateTime		End of the rental.	2026-09-28T09:00Z
PickupLocation	string(100)		Pickup point. Required.	Downtown Hub
RentalDuration	decimal(18,2)		Units (hours/days) charged.	2.00
BasePrice	decimal(18,2)		Price before discount.	3600.00
DiscountAmount	decimal(18,2)		Discount applied.	0.00
TotalAmount	decimal(18,2)		BasePrice - DiscountAmount.	3600.00
PromoCodeId	int?	FK?	-> PromoCode.Id (optional; SetNull).	1
PromoCode	PromoCode?	nav	Navigation to applied promo.	{ WELCOME20 }
Status	BookingStatus		Pending, Confirmed, Completed, Cancelled.	Confirmed
CreatedAt	DateTime		When booking was made (UTC).	2026-09-20T12:00Z
Payment	Payment?	nav	Navigation: 1-to-1 payment.	{ TXN-... }
Review	Review?	nav	Navigation: 1-to-1 review.	{ 5 stars }

4. Payment

Payment record tied one-to-one to a booking. Deleting the booking cascades to the payment.

Attribute	Data Type	Key	Stores / Description	Example
Id	int	PK	Auto-generated unique identifier.	5
BookingId	int	FK	-> Booking.Id (1-to-1, cascade).	10
Booking	Booking?	nav	Navigation to the paid booking.	{ DE-202609-1002 }
TransactionId	string(50)		Gateway transaction ref. Required.	TXN-20260920-90823
PaymentMethod	PaymentMethod		Card, UPI, NetBanking or Wallet.	Card
Amount	decimal(18,2)		Amount paid.	3600.00
Status	PaymentStatus		Pending, Successful or Failed.	Successful
PaidAt	DateTime?		When payment succeeded (nullable).	2026-09-20T12:05Z

5. Review

A rating and comment left after a completed booking. One-to-one with Booking; links to both the User and the Vehicle.

Attribute	Data Type	Key	Stores / Description	Example
Id	int	PK	Auto-generated unique identifier.	7
UserId	int	FK	-> User.Id (reviewer).	3
User	User?	nav	Navigation to the reviewer.	{ Priya }
VehicleId	int	FK	-> Vehicle.Id (reviewed vehicle).	4
Vehicle	Vehicle?	nav	Navigation to reviewed vehicle.	{ Honda City }
BookingId	int	FK	-> Booking.Id (1-to-1, cascade).	9
Booking	Booking?	nav	Navigation to the source booking.	{ DE-202608-1001 }
Rating	int		Star rating, range 1-5.	5
Comment	string(1000)		Review text. Required, 5-1000 chars.	Excellent self-drive experience!
CreatedAt	DateTime		When review was written (UTC).	2026-09-14T10:00Z

6. PromoCode

A discount coupon applied to a booking. Code is uniquely indexed. Discount is a percentage or a flat amount, valid within a date window.

Attribute	Data Type	Key	Stores / Description	Example
Id	int	PK	Auto-generated unique identifier.	1
Code	string(20)		Coupon code. Required, unique.	WELCOME20
DiscountType	DiscountType		Percentage or FlatAmount.	Percentage
DiscountValue	decimal(18,2)		20 (%) or a flat Rs. value.	20.00
MinBookingAmount	decimal(18,2)		Minimum base price to qualify.	1000.00
ValidFrom	DateTime		Start of validity window.	2026-08-21T00:00Z
ValidTo	DateTime		End of validity window.	2026-12-21T00:00Z
IsActive	bool		Whether the code is enabled.	true

7. PricingRule

Dynamic pricing rule (e.g. weekend surge). Optionally scoped to a vehicle type and/or a day of week; applies a multiplier to base price.

Attribute	Data Type	Key	Stores / Description	Example
Id	int	PK	Auto-generated unique identifier.	1
Name	string(50)		Rule label. Required.	Weekend Demand Surge (+15%)
VehicleType	VehicleType?		Optional type scope (null = all).	null
Multiplier	decimal(5,2)		Factor applied to base price.	1.15
DayOfWeek	DayOfWeek?		Optional day scope (null = any).	Saturday
IsActive	bool		Whether the rule is enabled.	true

8. Notification

An SMS-style message sent to a user. Deleting the user cascades to their notifications.

Attribute	Data Type	Key	Stores / Description	Example
Id	int	PK	Auto-generated unique identifier.	1
UserId	int	FK	-> User.Id (recipient, cascade).	2
User	User?	nav	Navigation to the recipient.	{ Rahul }
RecipientPhone	string(15)		Destination phone. Required.	9823456780
Message	string(300)		Message body. Required.	Welcome to DriveEase, Rahul!
Type	NotificationType		BookingConfirmation / Cancelled / PaymentAlert.	BookingConfirmation
IsSent	bool		Whether it was dispatched.	true
SentAt	DateTime		When it was sent (UTC).	2026-09-20T12:00Z

Enumerations Used

Enum	Values
UserRole	Customer (1), Administrator (2)
RentalType	Hourly (1), Daily (2)
VehicleType	Bike (1), Hatchback (2), Sedan (3), SUV (4), Luxury (5)
FuelType	Petrol (1), Diesel (2), Electric (3)
TransmissionType	Manual (1), Automatic (2)
VehicleStatus	Available (1), Booked (2), UnderMaintenance (3)
BookingStatus	Pending (1), Confirmed (2), Completed (3), Cancelled (4)
PaymentStatus	Pending (1), Successful (2), Failed (3)
PaymentMethod	Card (1), UPI (2), NetBanking (3), Wallet (4)
DiscountType	Percentage (1), FlatAmount (2)
NotificationType	BookingConfirmation (1), BookingCancelled (2), PaymentAlert (3)

Model Relationships

Relationships are configured in `ApplicationDbContext.OnModelCreating`. The table below lists each association, its cardinality, the foreign key, and the on-delete behavior.

Relationship	Cardinality	Foreign Key	On Delete
User → Booking	One-to-Many	Booking.UserId	Restrict
User → Review	One-to-Many	Review.UserId	Restrict
User → Notification	One-to-Many	Notification.UserId	Cascade
Vehicle → Booking	One-to-Many	Booking.VehicleId	Restrict
Vehicle → Review	One-to-Many	Review.VehicleId	Restrict
Booking → Payment	One-to-One	Payment.BookingId	Cascade
Booking → Review	One-to-One	Review.BookingId	Cascade
PromoCode → Booking	One-to-Many (optional)	Booking.PromoCodeId (nullable)	SetNull

Reading the delete behaviors

Restrict — you cannot delete a User or Vehicle while it still has bookings or reviews; those must be removed first. This protects historical records.

Cascade — deleting the parent automatically deletes the children. Deleting a Booking removes its Payment and Review; deleting a User removes their Notifications.

SetNull — deleting a PromoCode does not delete bookings that used it; their PromoCodeId is simply set to null, preserving the booking history.

Unique Indexes

The following columns are enforced as unique at the database level: **User.Email**, **Vehicle.LicensePlate**, **Booking.BookingReference**, and **PromoCode.Code**.

Text Relationship Map

```
User (1) ■■■■< (many) Booking >■■■■ (1) Vehicle
|
|           | |
|           | ■■■■(1:1)■■■ Review ■■■■(many:1)■■■> Vehicle & User
|           | ■■■■(1:1)■■■■ Payment
|
| ■■■■(1:many)■■■ Review
| ■■■■(1:many)■■■ Notification

PromoCode (1) ■■■■< (0..many) Booking [optional, SetNull]
PricingRule : standalone (no FK; matched by VehicleType / DayOfWeek at runtime)
```